

Project Planning Phase
Project Planning Template (Product Backlog, Sprint Planning, Stories, Story points)

Date	3 July 2026
Team ID	
Project Name	Smart Agricultural Production Optimization Engine
Maximum Marks	5 Marks

Product Backlog, Sprint Schedule, and Estimation (4 Marks)

Use the below template to create product backlog and sprint schedule

Sprint	Functional Requirement (Epic)	User Story No.	User Story / Task	Story Points	Priority	Team Members
Sprint-1	Farmer Input Management	US-1	Enter Nitrogen (N), Phosphorus (P), and Potassium (K) values.	3	High	Team
Sprint-1	Farm & Soil Data Collection	US-2	Enter soil pH value.	2	High	Team
Sprint-1	Farm & Soil Data Collection	US-3	Enter temperature, humidity, and rainfall values.	3	High	Team
Sprint-1	Input Validation	US-4	Validate soil and environmental input values before prediction.	3	High	Team
Sprint-2	Machine Learning Crop Recommendation	US-5	Generate crop recommendation using the trained machine learning model.	5	High	Team
Sprint-2	Prediction Results Display	US-6	Display the recommended crop to the farmer.	2	High	Team

Sprint-2	User Interface	US-7	Provide a simple and user-friendly interface for data entry and prediction.	3	Medium	Team
Sprint-3	Data Analysis & Preprocessing	US-8	As a farmer, I want the system to process my input data correctly so that I receive an accurate crop recommendation.	3	Medium	Team

Project Tracker, Velocity & Burndown Chart: (4 Marks)

Sprint	Total Story Points	Duration	Sprint Start Date	Sprint End Date (Planned)	Story Points Completed	Sprint Release Date (Actual)
Sprint-1	11	2 Days	26 Jun 2026	27 Jun 2026	11	27 Jun 2026
Sprint-2	10	2 Days	28 Jun 2026	29 Jun 2026	10	29 Jun 2026
Sprint-3	3	2 Days	30 Jun 2026	01 Jul 2026	3	01 Jul 2026

Velocity:

Velocity = Total Story Points Completed ÷ Number of Sprints

Total Story Points = 24

Number of Sprints = 3

Velocity = $24 \div 3 = 8$ Story Points per Sprint

Burndown Chart:

